GLAUCOMA

BY

CHEGE MARGARET

REVIEW-
-

INTRAOCULAR PRESSURE

- Intraocular pressure normal = 12 – 20 mmHg. It is largely depended upon the structures within the eyeball, but especially the **aqueous** humour.

Factors that maintain normal IOP:

-Rate of formation of Aqueous humour

-Rate of drainage of AH through trabecular mess

- Pressure in the episcleral veins(Sinus Venosus Sclerae) in which canal of Schlemm drains

- Elasticity of the cornea and sclera- remain constant

The Aqueous Humour

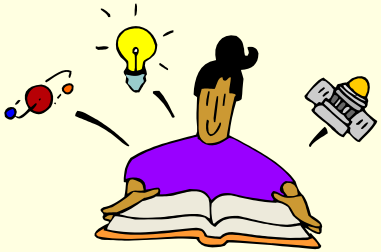
- Is transparent fluid.
- Produced by the epithelium cells of the ciliary body into the posterior chamber, between the lens and the iris.
- Circulates through the pupil to the anterior chamber.
- It is then absorbed into Sinus Venosus Sclerae after passing through the canal of schlemn at the angel of filtration (where the posterior surface of the cornea joins the iris).
- From the Sinus Venosus Sclerae it enters the general circulation.
- The trabecular muscle at the angle of filtration controls passage of humour to the canal of schlemn

The aqueous fluid varies according to:

- Rate of production
- Rate filtration .

If excessive production – glaucoma (rare)

GLAUCOMA



GLAUCOMA – What is it?

- **The Intraocular Pressure becomes pathologically high- can lead to blindness**

Why ? <u>Obstruction to Aqueous drainage from Anterior chamber</u>

Classifications Of Glaucoma

- *1.Open-angle glaucoma*
- *2. Closed angle glaucoma*.
- 3. Secondary
- 4. Congenital

Pathophysiology of Glaucoma

- The mechanism of visual loss is ganglion cell atrophy …The Optic nerve becomes atrophic
- The Iris and the Ciliary body also become atrophic

NB In acute closed-angle glaucoma Intraocular Pressure (IOP) may rise to 60-80mmHg resulting in ischaemic damage to Iris, corneal oedema and Optic nerve damage

open angle glaucoma

- →Relationship between the Iris root and the trabecular mess= normal.

Cause ♣→Obstruction to the aqueous outflow due to defect in the drainage mechanism(the block is in juxtacanalicular trabecular tissues → whereby collagen content of the trabecular fibres become increased blocking the pores of meshwork

- Most common→ 90% of cases present with this type. Usually hereditary

♣ Haemorrhage (clot) of the passage can also cause this type

open angle glaucoma → clinical features

- 1. Initially it is asymptomatic No pain. No redness. No irritation. Then the patient is slowly losing sight and requires change of glasses frequently.
- 2.Then loss of visual field. → Visual field is the entire areas that can be seen without shifting the eye. He sees only what is in front of him and not what is sideways.

open angle glaucoma → clinical features cont……

- 3. Cupping of the optic disc. There is thrombosis of the retinal veins. They seem to be lying deep inside the disc, due to direct pressure on the optic nerve. (Thro fundoscopy)

open angle glaucoma diagnosis

- Repeated Tonometry - IOP
- Fundoscopy – Cupped Optic
- Visual Fields↓

open A.G MANAGEMENT

- Medical (drugs) treatment to control the pressure is the treatment of choice

The patient is put on:-

- 1.Pilocarpine eye drops 0.25% - 0.5% 2 hourly.

Facilitates aqueous out flow. It is miotic. The smaller the pupil the thinner the iris, therefore giving more space for drainage of aqueous at angle of anterior chamber. It also reduces the formation of aqueous humour

♦Eserine sulphate 1% - ½ hourly. May also be used. (It is more powerful) but usually avoided because it is irritating..

Medical (drugs) treatment….cont..

- 2. Adrenaline 1% eye drops (here it is a vaso constrictor) it decreases rate of aqueous formation.

  or

- Timolol (beta blocker) 0.25% - 0.5%.-suppress aqueous production

- 3. Diamox tabs 250-500mgs tds x 3/7-diuretic effect

NB These drugs may be used alone or in combination

If no improvement, ON Medical Treatment then surgery = TRABECULECTOMY.

TRABECULECTOMY.

- In this operation the acqueous humour is made to drain through sclera

Pre operative prep. for TRABECULECTOMY.

- General pre operative care……
- Patient is reassured.
- patient explained that surgery will not improve sight but will prevent it becoming worse Conjuctival swab for C/S..
- Any infection, cough, constipation is treated before surgery . Better diet.

Post operative care following TRABECULECTOMY

- General post operative care……..
- The operated eye to remain covered for 5/7 .
- Patient to lie on back with as many pillows as he/she likes.
- To be fed in bed 2/7.
- Conjuctiva stitch on 5th day.
- Clean eye gently tds with N/saline and check for any abnormality.
- On discharge patient advised against sudden heavy exertion. To be followed closely in clinic.

Closed angle glaucoma

Common in women – ages 50 – 60 years

Cause

- Shallow anterior chamber. There is obstruction to the drainage areas by the base of the Iris. The iris tissue narrows the anterior chamber. Thus pressure increases in A. chamber.

Closed angle glaucoma- Clinical Features

Of sudden onset.

- Blurring vision.
- Corneal oedema
- May progress to an attack of acute glaucoma where by the patient presents with:-
- Severe eye pain and vomiting.
- Very red eye
- Pupil fixed and dilated
- Rapid loss of vision

NB: Should be treat as emergency

MANAGEMENT of C.A.G

If acute symptoms develop Then:-

- Patient started on I.V manitol 150 mls 60 drops per minute.
- I.V Diamox 500 mgs diluted in 10 cc Normal saline – stat then oral 250 mgs 250 mgs tds to control pressure.
- Pilocarpine drops 0.25% - 0.5% 1 – 2 hourly. Patient is monitored closely – i.e. visual fields & IOP
- To prevent another attack – surgery – **Peripheral Iridectomy**

Peripheral Iridectomy

- This operation is done after the pressure has been reduced (lowered) by medical therapy.
- It is advisable to do the same operation on the other eye as a prophylactic measure.
- Specific pre operative and post operative as for tuberculectomy

Secondary Glaucoma

- Caused or secondary to other eye diseases e.g. trauma, iritis

Treatment

- Treat the cause.
- Control pressure by Diamox tabs 250 mgs tds
- Pilocarpine eye drops. Then the patient is followed by closely in the clinic as may need surgery

Congenital Glaucoma (BUPHTHALMOS)

Cause

- Maldevelopment of Trabecular muscle or canal of schlemn so that drainage system is poor.

Congenital Glaucoma -Clinical Features

- Infant has large, grey eyes.
- Photophobia: (he protects his eyes from light).
- Usually bilateral.
- Deep anterior chamber → The corneal diameter is larger than normal.

Congenital Glaucoma-Management

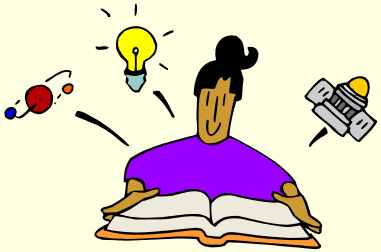
- Surgery

Either

- **Goniotomy**

A needle is inserted and rotated into an angle of filtration to widen or open the blockage.

OR

- **Trabeculectomy**

THANK YOU